

HYDROLOGIC ANALYSIS AND DESIGN

FOURTH EDITION



RICHARD H. McCUEN

Hydrologic Analysis and Design

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Hydrologic Analysis and Design

Fourth Edition

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Preface

Hydrologic Analysis and Design is intended for a first course in hydrology. It introduces the student to the physical processes of the hydrologic cycle, the computational fundamentals of hydrologic analysis, and the elements of design hydrology. Although the sections of the book that introduce engineering design methods are intended for the engineering student, the concepts and methods will also be of interest to, and readily understood by, students in other disciplines including geology, geography, forestry, and planning. The material is appropriate for undergraduate and graduate students as well as for the practitioner interested in reviewing the fundamentals of hydrology. While computers are widely used in the practice of hydrology, knowledge of computer hardware, software, or programming languages is not necessary to use this book; however, computer literacy will certainly facilitate problem solving.

A number of factors influenced the structure of the book. Since many undergraduates get internships or post-graduate jobs in land development, the book focuses on related topics. However, the book will also be of value to those going directly to graduate school. The background in this book will provide a strong basis for graduate courses in hydrologic modeling and the physical processes. Since hydrologic design and analysis, especially as it relates to land development, begins with rainfall, it is logical to start the discussion with precipitation; thus, Chapter 2 presents the concepts related to rainfall that are needed in most of the other chapters. The level of flooding depends on watershed characteristics as well as the rainfall, so Chapter 3 is devoted to watershed factors that influence runoff rates and volumes. Chapter 4 presents concepts related to frequency analysis, as this topic addresses methods used in the development of peak discharge rates. Chapter 5 introduces methods for estimating design discharge rates, as many design problems only require an estimate of a peak rate. Chapter 6, which is the longest chapter, is devoted to hydrologic design, including stormwater management, wetlands, and a few best management practices. Most design work should be done with hydrograph methods, so Chapter 7 is devoted to the use of hydrographs in design. Most hydrologic design work is made on nonhomogeneous watersheds, that is, differences in land use/land cover, soils, and slopes; therefore, Chapter 8 is devoted to hydrologic routing, both channel routing and storage facility routing. In northern climates and in high-elevation semidesert areas, snowmelt runoff is a principal source of water, for both water supply and flooding; therefore, Chapter 9 provides an introduction to water yield and snowmelt runoff. Chapter 10 provides a brief overview of evaporation, which is a principal loss. Chapter 11 discusses erosion and sedimentation, which will provide a good lead-in to Chapter 12, which includes topics related to the basics of water quality modeling. Groundwater topics are addressed in Chapter 13. A series of short appendices are used to provide a brief overview of general topics.

Because of the diverse ways that hydrology can be taught, flexibility was a criterion used in structuring the book. The need for flexibility led to a book comprising 13 chapters. To the extent possible, I tried to make the chapters self-contained so that the order in which the topics are presented is not constrained by the book's structure. Given the orientation of the book toward design, I carefully examined the types of design problems that were to be emphasized. Land development and urban hydrology are very important aspects of hydrology, and therefore, many of the topics that are used in such design work were placed at the beginning of the book.

Material for most topics in the fundamentals of hydrology can be divided into four areas: physical processes, measurement and collection of data, model conceptualization and data analysis, and design or solution synthesis. A basic knowledge of the physical processes is necessary to properly analyze data or perform design work. Thus, I have tried to provide a discussion of the physical processes associated with the components of the hydrologic cycle. The discussion is limited to the dominant factors, especially those measurable characteristics that are used in design. The measurement and collection of data is one area of hydrology where many hydrologists are employed. This topic is not emphasized here because the book is intended to be a student's first exposure to topics in hydrology. The title, "Hydrologic Analysis and Design," indicates that the emphasis has been placed on the last two of the four areas identified, namely model conceptualization and data analysis, and design or solution synthesis. In most chapters, a brief section that discusses analysis and synthesis is included. Emphasis has been placed on analysis and synthesis because I believe that knowledge of these subjects is most important for the beginning student. Before anyone performs a hydrologic design, the person should fully understand both the theoretical origin of and any empirical basis for the design method. The last two of the four topics are the ones that students usually find most interesting; thus, by emphasizing these areas I hope that the book will motivate students toward hydrologic practice. I believe that detailed discussions of the physical processes and data collection should follow the introductory discussion of the methods of analysis and design. This belief shaped the emphasis that is placed on the topics covered.

Educational programs in engineering and the sciences are often criticized as being too theoretical. Practitioners often complain that recent graduates of engineering programs cannot perform basic design computations. *Hydrologic Analysis and Design* has been structured to overcome this problem. While the text is not intended to be a design manual, a number of design methods are included. It is important to recognize that every design problem has alternative methods that can be used. Not all of the alternatives can be presented within the pages of this book, so I tried to select the ones that best illustrated the physical processes that underlie the problem. Although all of the details of these design methods are not given, the material provided should demonstrate to the student that design computations are not difficult and that the design methods have a basis in the fundamental concepts of the hydrologic cycle.

Students of engineering hydrology and the hydrologic sciences often fail to recognize the interconnection between topics discussed in an introductory

course in hydrologic analysis and design. Engineering design requires knowledge of watershed characteristics, rainfall, frequency concepts, and the processes that control runoff. In an attempt to alleviate this problem, I have developed a series of projects. I have used these projects for decades with much success. While a few students complain of the effort required, almost all of them agree that the project approach is both interesting and educational. The “real world” flavor of the project approach to learning hydrology definitely serves to inspire interest in hydrologic analysis and design. I highly recommend supplementing the lecture format with the project approach.

In addition to the project approach to learning, I have included two other items that are different from most books on hydrologic analysis and design. Each chapter includes a series of multiple-choice review questions. These are designed to encourage the student to understand that the qualitative concepts of hydrology are just as important as the quantitative design methods. If these review questions are assigned at the end of the discussion of a chapter, they can also serve to tie together the different concepts discussed in the chapter. I have included material on ethical issues in hydrology (the last few sections of Chapter 1). The purpose of the material is to introduce the student to another side of hydrologic practice, namely the societal or value element that accompanies every hydrologic analysis and design. While the book is devoted to the quantitative aspects of hydrology, the practitioner should always give consideration to the societal effects of a project. Those involved in the practice of hydrologic analysis and design do not work in a social vacuum: The presented material will make this point to the student. Students always enjoy the lecture that is associated with Chapter 1, and some former students have told me that it was the most important lecture. Additionally, each chapter includes a discussion question that poses a situation that requires knowledge of values and value decision making.

New to the Fourth Edition. In progressing from the third edition to this edition, the primary changes have been to streamline the organization of the chapters and reduce the length of the book. While topics have been deleted, the topics were secondary to the overall educational objectives of the book. For example, the large section that discussed the inputs to the TR-20 program was eliminated as the material is readily available in the online user’s manual. The chapter on statistical methods was eliminated as the material is generally taught in other introductory courses. Thus, I do not believe that these deletions have detracted from the ability of the book to be a primary source of knowledge on hydrologic analysis and design. In summary, improvement by subtraction!

Additions have been made to provide new knowledge about important topics, including:

- Design procedures related to best management practices have been included.
- A new philosophy for computing times of concentration forms the basis for a revised section on a topic that is central to many hydrologic designs.
- A method for computing runoff curve numbers that depend on the storm duration has been developed and included.

- A section on the design of constructed wetlands is included.
- A discussion of the energy budget approach to snowmelt runoff is now included.
- A new method of adjusting the Rational method runoff coefficient for slope and return period is presented.
- Hydrologic design methods are not highly accurate so discussions of the effect of input inaccuracy have been added to several topics.
- Underlying assumptions of design methods, such as the Rational method, have been added to help the user decide if the method is appropriate for a specific type of design.
- The calculation of stage–storage–discharge relations has been improved.
- Additional end-of-chapter exercises have been added.
- Brief appendices that outline specific topics have been added.

I want to acknowledge the help of the following people who reviewed drafts of selected chapters of the first edition: Henry Anderson, Mark Hawley, David Murray, Walter Rawls, and Peter Waldo. Their comments contributed to the success of the first edition. I also greatly appreciate the helpful suggestions of the following reviewers of a prospectus of this fourth edition: Nikolaos Katopodes, University of Michigan; Tyler Smith, Clarkson University; Tony Cahill, Texas A&M University; Jy Wu, University of North Carolina at Charlotte; Gerald Kauffman, University of Delaware; Paolo Scardina, Virginia Polytechnic Institute and State University; and Jeffrey Niemann, Colorado State University. Of course, I remain responsible for all errors or omission of topics. I also want to acknowledge the many students who have read drafts of various sections of the manuscript. The final product is much improved because of their efforts. And finally, but most importantly, I am very grateful for those involved in the production of this edition, especially those from the publisher (Erin Ault, Scott Disanno, Holly Stark and Camille Trentacoste) and the contractor (Rosalind Shahuna and Mohanarengan Dilli). They were a pleasure to work with and their responsiveness and efficiency were outstanding.

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Hydrologic Analysis and Design

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Stormwater infrastructure can be an aesthetic asset to a community.

1 Introduction to Hydrology

CHAPTER OBJECTIVES

- Define hydrology.
- Introduce the hydrologic cycle.
- Develop a conceptual understanding of society's effect on the hydrologic cycle.
- Distinguish between hydrologic analysis and synthesis.
- Introduce the linear-storage equation.
- Demonstrate that hydrologic practice involves more than just solving technical design problems; hydrologists do not work in a social vacuum.
- Introduce the fundamental components of professionalism and place them in the context of hydrologic practice.

- Identify human and social values that are important for the hydrologist to consider.
- Provide a framework for understanding and solving value dilemmas in a professional setting.

1.0 NOTATION

b_o = model coefficient (intercept)

b_1 = model coefficient (slope)

E = evaporation

I = inflow

O = outflow

S = storage

t = time

T = temperature

1.1 HYDROLOGY: THE STUDY OF WATER

The word *hydrology* combines the Greek word *hudōr*, which means “water,” and the term *-logy*, which designates “a study of.” It also has origins in the New Latin word *hydrologia*. More specifically, the general word *hydrology* refers to the scientific study of water and its properties, distribution, and effects on Earth’s surface, soil, and atmosphere. The study of water can mean different things to different professions. To a chemist, a water molecule is a stable chemical bond of two atoms of hydrogen and one atom of oxygen; the chemist is interested in the properties of water and its role in chemical reactions. The climatologist is interested in the effect of the water stored in the soil and lakes on climatic processes. To those involved in the design of hydraulic machinery, the study of the properties of water concentrates on the forces exerted by water in a dynamic state. To the mechanical engineer, the properties of water in the form of steam can be important. The groundwater hydrologist is interested in the movement of water in transporting pollutants. Even geographers and historians may be interested in water, at least in terms of how its availability and accessibility have shaped development and culture. However, our interest herein is in the narrow field of hydrologic engineering analysis and design. Engineering hydrology encompasses those aspects of hydrology that relate to the design and operation of engineering projects for the control and use of water. Aspects that relate to our nation’s infrastructure are of special interest because of the current decaying state of our infrastructure. Whereas the primary interest is to the engineer, others, such as planners, environmentalists, water managers, and meteorologists, may find the knowledge of hydrologic engineering to be of interest.

Earth’s atmosphere, oceans, ice masses, lakes, rivers, streams, and soil contain more than 50 billion cubic feet of water. In spite of this abundance, problems are created by either too much or too little water at a given location (i.e., problems are caused by the spatial variation of water). For example, people living in southern California and other areas of the arid southwest show concern over the lack of an inexpensive source of water supply. Problems also result from variations in the time distribution of water. An overabundance of water at one time or an undersupply at

other times can have serious consequences to both agriculture and manufacturing and can inconvenience the urban public. Occasional flooding is a problem to homeowners and to entire cities. Crops do not grow at the optimum rate when soil is either too wet or too dry. Manufacturing operations require a consistent water supply over time for a variety of purposes, such as to provide cooling water and assimilate wastes. Thus, although Earth's total volume of water may be adequate to meet all needs, problems are created by variations in both the spatial and temporal distributions of water availability. Extreme problems, including life-threatening situations, can result from extreme variations in either the spatial or temporal distribution of water or both.

In an attempt to overcome the problems created by these variations in the temporal and spatial variations in water availability, engineers and hydrologists attempt to predict water availability. These predictions are used in the evaluation of alternative means of preventing or solving problems. A number of factors contribute to the ineffectiveness of these engineering designs. First, the occurrence of rainfall cannot be predicted with certainty—that is, it is not possible to predict exactly how much rain will occur in one time period (e.g., a day, month, or year). The uncertainty of extreme variation in rainfall amounts is even greater than the uncertainty in the rainfall volumes occurring in the more frequent storm events. It is difficult to design engineering works that will control the water under all conditions of variation in both the time and spatial distributions. Second, even if we had perfect information, the cost of all of the worthwhile projects needed to provide the optimum availability of water is still prohibitive. Therefore, only the most efficient and necessary projects can be constructed. Third, hydrologic processes such as rainfall and runoff are very complex, and a complete, unified theory of hydrology does not exist. Therefore, measurements of observed occurrences are used to supplement the scant theoretical understanding of hydrologic processes that exists. However, given the limited records of data, the accuracy of many engineering designs is less than we would like. These three factors (hydrologic uncertainty, economic limitations, and lack of theory and observed data) are just some of the reasons that we cannot provide solutions to all problems created by undesirable variations in the spatial and temporal distributions of water.

In spite of the inherent uncertainty in precipitation, the economic constraints, and the bounds on our theoretical understanding of hydrometeorological processes, solutions to the problems that are created by the temporal and spatial variations in water availability must be provided. Estimates of hydrologic quantities such as stream flow are required as input to engineering designs, which represent the engineer's attempt to solve the problem. All engineering designs should, at the minimum, be rational. An understanding of the physical processes is a necessary prerequisite to the development of rational designs.

1.2 THE HYDROLOGIC CYCLE

The physical processes that control the distribution and movement of water are best understood in terms of the hydrologic cycle. Although a real beginning or ending point of the hydrologic cycle cannot be identified, we can begin the discussion

with precipitation. For the purposes of this discussion, we will assume that precipitation consists of rainfall and snowfall. A schematic of the hydrologic cycle for a natural environment is shown in Figure 1.1. Rain that falls on Earth may enter a water body directly, travel over the land surface from the point of impact to a watercourse, or infiltrate into the ground. Some rain is intercepted by vegetation; the intercepted water is temporarily stored on the vegetation until it evaporates back to the atmosphere. Some rain is stored in surface depressions, with almost all of the depression storage infiltrating into the ground. Water stored in depressions, water intercepted by vegetation, and water that infiltrates into the soil during the early part of a storm represent initial losses. The loss is water that does not appear as runoff during or immediately following a rainfall event. Water entering the upland streams travels to increasingly larger rivers and then to seas and oceans. The water that infiltrates into the ground may percolate to the water table or travel in the unsaturated zone until it reappears as surface flow. The amount of water stored in the soil determines, in part, the amount of rain that will infiltrate during the next storm event. Water stored in lakes, seas, and oceans evaporates back to the atmosphere, where it completes the cycle and is available for rainfall. Water also

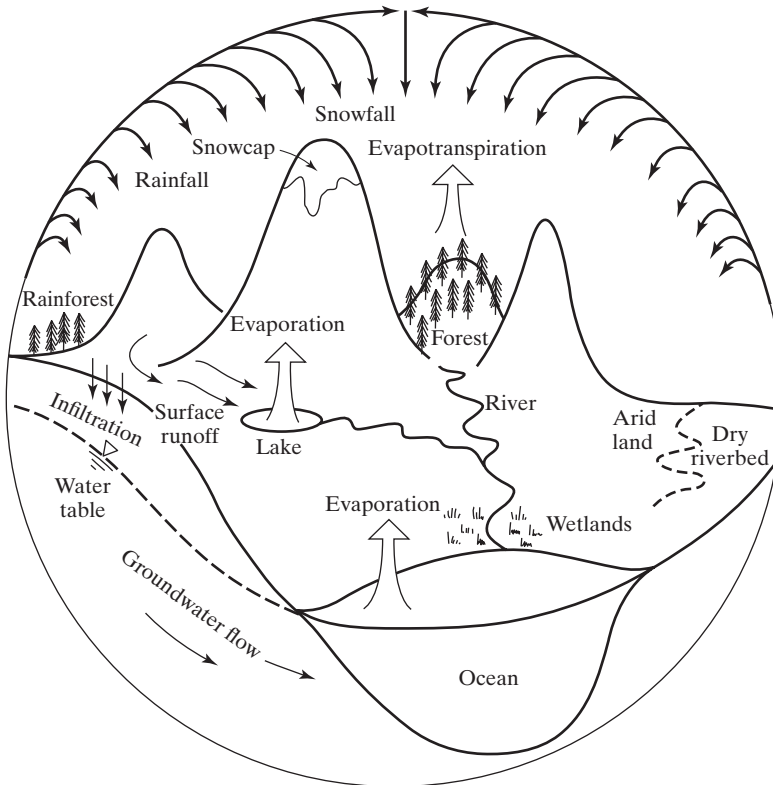


Figure 1.1 Hydrologic cycle of a natural environment.

evaporates from soil devoid of vegetation. Rain that falls on vegetated surfaces may be intercepted—however, after the storage that is available for interception is filled, the water will immediately fall from the plant surfaces to the ground and infiltrate into the soil in the same way that water falling on bare ground infiltrates. Some of the water stored in the soil near plants is taken up by the roots of the vegetation and subsequently is passed back to the atmosphere from the leaves of the plants; this process is called *transpiration*.

Although it may appear that the hydrologic cycle of a natural environment is static, it is important to recognize that the landscape is constantly undergoing a transformation. High-intensity storms cause erosion of the land surface. Flood runoff from large-volume storms causes bankfull and high-velocity flows in streams with the potential for large amounts of channel erosion. During periods of extreme drought, the perimeter of desert lands may increase. Forest fires caused either by natural means such as electrical storms or by the carelessness of human beings cause significant decreases in available storage and decrease surface roughness, both of which contribute to increases in surface-runoff rates and volumes, as well as surface erosion. When mud flows are a potential problem, forest fires consume the vegetation, which contributes significantly to the increased production of debris. In many parts of the world, the largest floods result from the rapid melting of snow; such events can be as devastating as floods produced by large rainfall events. Flooding accompanying hurricanes and monsoons also can cause significant changes to the landscape, which by itself affects runoff rates and volumes from storms occurring long after the hurricane or monsoon. In summary, even in a natural environment, rainfall and runoff cause major changes in the watershed.

As the population of the world has increased, changes to the land have often been significant, with major changes to the runoff characteristics of a watershed as a result. Land clearing for agricultural development increases the amount of exposed soil, with obvious decreases in the protective covering of the natural vegetation. This loss of protective covering decreases the potential for infiltration, increases surface runoff, and often results in significant soil losses. Over the last two centuries, urbanization has caused significant changes to the landscape surrounding these urban centers. Urbanization has had significant effects on the processes of the hydrologic cycle for watersheds subject to urban development. Clearing of the land has reduced vegetation and therefore the availability of interception storage; grading of land surfaces reduces the available volume of depression storage. Impervious surfaces reduce the potential for infiltration and the resulting recharge of groundwater storage. Impervious surfaces are also less rough than natural surfaces and thus offer less resistance to runoff; this change in roughness can increase runoff velocities and surface erosion. These changes to the processes of the hydrologic cycle, which are shown schematically in Figure 1.2, cause significant changes in runoff characteristics. The reduced storage results in increased volumes of surface runoff. The reduced surface roughness decreases the travel time of runoff. The reductions in both storage and travel time result in increased peak rates of runoff, which increase both flood damages due to overbank flows and channel erosion.

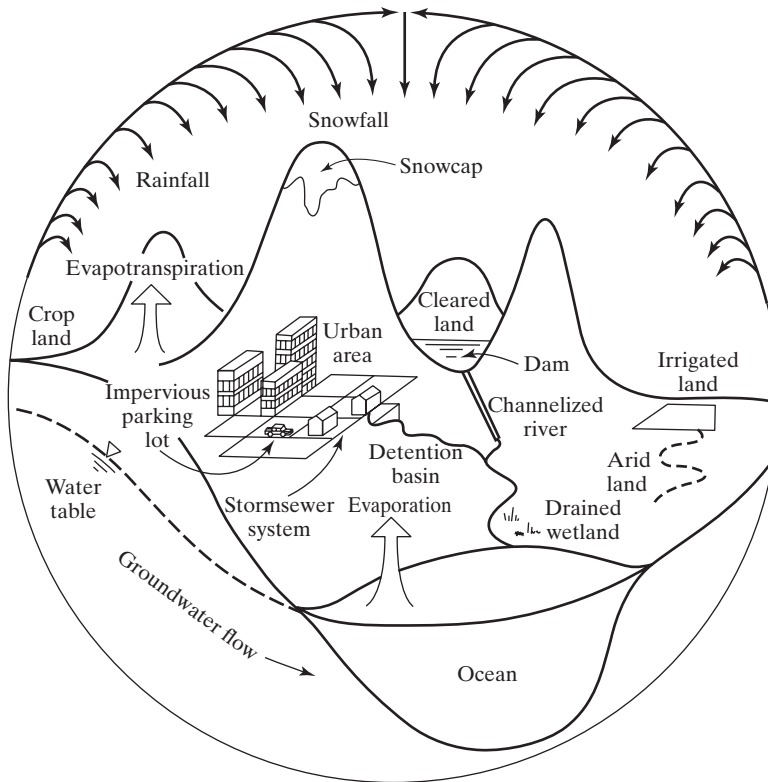


Figure 1.2 Hydrologic cycle of a developed environment.

In an attempt to compensate for lost natural storage, many localities require the replacement of lost natural storage with human-made storage. While the storm water detention basin is the most frequently used method of storm water management, other methods are used, such as infiltration pits, rooftop and parking lot storage, and porous pavement. The current trend is to design newly developed areas to have minimal hydrologic effects. These engineering works do not always return the runoff characteristics to those that existed in the natural environment. In fact, poorly conceived methods of control have, in some cases, made flood-runoff conditions worse.

EXERCISES

- 1.2.1** Identify the elements of the hydrologic cycle that are important for a single residential lot.
- 1.2.2** When a new roadway is built, what changes of the hydrologic cycle are made to that piece of land?

1.3 HYDROLOGIC DESIGNS

The average American drives about 25 miles per day. While many of these miles involve travel on the same route, say to or from school or a job, the individual traveler encounters a surprisingly large number of engineering designs where hydrologic analyses were required. In fact, hydrology was considered in the design of the very 25 miles of highway covered in an average day. Failure to consider drainage of the highway sub-base can lead to premature failure of the highway. In addition to sub-base design, hydrologic analyses are required for the design of culverts (e.g., a pipe that crosses under a road or embankment), surface drainage inlets, and bridges that cross over rivers and streams. It should be evident that those involved in highway design must understand the basic concepts of hydrologic analysis since the design of every mile of highway requires consideration of the fundamental concepts of hydrology. Other elements of transportation systems require hydrologic analyses. Those involved in the design of parking lots, airport runways and aprons, rapid-mass-transit lines, and train rights-of-way must give just as much consideration to the proper drainage of storm runoff as those involved in the design of highways.

Those involved in the design of transportation facilities are not the only ones who must consider the natural passage of water resulting from storm events. Anyone involved in land development and the construction of homes, as well as in the construction of commercial, industrial, and institutional buildings, must consider storm runoff. Obviously, those involved in home design must provide gutters and downspouts. Buildings in commercial and industrial developments also require roof drainage.

Many other hydrologic analyses are required in building construction. When clearing land for development, it is important to provide sediment-control facilities to ensure that eroded soil does not enter into waterways and wetlands. Sediment control depends on the area of the land being cleared, the amount of rainfall that can be expected during the period where the soil will be exposed to rainfall impact, and site characteristics such as the slope and soil type. In addition to hydrologic considerations during the land development stage, site development must consider drainage patterns after development.

Site development usually results in significant increases in impervious surfaces, which results in increased surface-runoff rates and volumes. In many localities, storm water control facilities are required. In the upper reaches of a site, swales can be used to move water away from buildings and transportation facilities. Concentrated runoff from swales may enter gutters and drainage ditches along roadways where the runoff may drain into highway inlets or small streams. At many sites where land development has resulted in large amounts of imperviousness, on-site detention basins may be required to control the increased runoff. The design of storm water detention basins requires knowledge of routing of water through the hydraulic outlet structure, as well as knowledge about surface runoff into the detention basin. The design must consider meteorological factors, geomorphological factors, and the economic value of the land, as well as human-value considerations such as aesthetic and public safety aspects of the design. The

design of a storm water detention basin should also consider the possible effects of inadequate maintenance of the facility.

The hydrologic designs discussed in the preceding paragraphs are based primarily on rainfall and the resulting surface runoff. Dams and the water stored in the reservoirs behind the dams provide many benefits, such as power generation, recreation, flood control, irrigation, and maintenance of low flows for water-quality control. In addition to estimating the volume of inflow into the reservoir, dam design requires assessment of the evaporation losses from the reservoir. For reservoirs with large surface areas, evaporation losses can be significant. Failure to consider evaporation losses during the design could result in overestimating the water that would be available for the aforementioned purposes. Thus, failure to understand the processes of the hydrologic cycle may render the design inadequate.

A design engineer should fully understand the theoretical and empirical basis for any design method used. However, it is equally important for the design engineer to reflect on the accuracy of the inputs to the design. In a course on the fundamentals of fluid dynamics, the velocity input to Bernoulli's equation is generally viewed as a precise value. In reality, velocity estimates are quite uncertain. The design engineer should have a sense of the inaccuracy and appreciate the potential effect on a design and its safety. This idea of hydrologic inputs being imprecise will be discussed relevant to some of the design methods.

EXERCISES

- 1.3.1 Obtain a dictionary definition of the word *design*. Discuss the concept of hydrologic design as it relates to the installation of a small pond on a farm, where the pond will be used for both irrigating crops during dry periods and watering livestock.
- 1.3.2 Using the dictionary definition of the word *design*, discuss the concept of hydrologic design as it would apply to the rooftop of a large shopping mall.

1.4 ANALYSIS VERSUS SYNTHESIS

Like most of the basic sciences, hydrology requires both analysis and synthesis to use the fundamental concepts in the solution of engineering problems. The word *analysis* is derived from the Greek word *analusis*, which means “a releasing,” and from *analuein*, which means “to undo.” In practical terms, *analysis* means “to break apart” or “to separate into its fundamental constituents.” Analysis should be compared with the word synthesis. The word *synthesis* comes from the Latin word *synthithenai*, which means “to put together.” In practical terms, *synthesis* means “to combine separate elements to form a whole.”

Because of the complexity of most hydrologic-engineering design problems, the fundamental elements of the hydrologic sciences cannot be used directly. Instead, it is necessary to take measurements of the response of a hydrologic process and analyze them in an attempt to understand how the process functions. Quite frequently, a model is formulated on the basis of the physical concepts that underlie the process, and the fitting of the model with the measurements provides the basis

for understanding how the physical process varies as the input to the process varies. After the measurements have been analyzed (taken apart) to fit the model, the model can be used to synthesize (put together) design rules—that is, the analysis leads to a set of systematic rules that explain how the underlying hydrologic process will function in the future. We recognize that the act of synthesizing is not a total reproduction of the original process; instead, it is a simplification. As with any simplification, it will not provide a totally precise representation of the physical process under all conditions. But, in general, it should provide reasonable solutions, especially when many designs based on the same design rules are considered.

It should be emphasized that almost every hydrologic design (or synthesis) is preceded by a hydrologic analysis. Most often, one hydrologic analysis is used as the basis for many hydrologic designs. But the important point is that the designer must understand the basis for the analysis that underlies any design method; otherwise, the designer may not apply the design procedure in a way that is compatible with the underlying analysis. While a design method can be applied without knowing the underlying analysis, it is best when the design engineer fully understands the analysis that led to development of the design rules. Anyone can substitute the values of input variables into a design method. But when a design is used under circumstances for which it was not intended, inaccurate designs can result. Without further consideration, the engineer uses these estimates as the basis of the design.

1.4.1 Water Budget Analysis and Synthesis. A simplified example may help illustrate the concepts of analysis and synthesis. Let's assume that an engineer has the task of designing a reservoir, and the design requires estimates of the water loss by evaporation from the reservoir. While the design period will be from June 1 to July 31, the engineer is assigned the project just before October 1, with a completion date of April 15 of the following year.

Recognizing his lack of experience in making hydrologic evaluations, the engineer correctly decides to hire a hydrologist to perform an analysis to provide a means of making estimates of evaporation. Recognizing that temperature is an important determinant in the evaporation process, the hydrologist decides to develop a design procedure based on a model relating evaporation to temperature. Since meteorological data-collection stations are not available in the region of the design, the hydrologist decides to collect data of the monthly evaporation from a nearby lake and compute the mean monthly temperature. The following data are the result of the data-collection effort:

Month	T (°F)	E (in.)
October	70	4.3
November	55	3.3
December	51	3.2
January	50	2.9
February	53	3.1
March	60	3.6

An examination of the literature suggests that evaporation can be related to temperature using a power model $E = b_0 T^{b_1}$ with the values of b_0 and b_1 computed using regression analysis. The analysis of the preceding data leads to the model $E = 0.04368 T^{1.0792}$ with a correlation coefficient of 0.98, which suggests that the model should be accurate. To summarize, the analysis problem involved (1) deciding on the variables to use, (2) deciding on the form of the model, (3) collecting the matrix of E and T data, and (4) fitting the model to the data and evaluating the expected accuracy of the design model. The hydrologist provides the engineer with the foregoing model.

Given the results of the preceding analysis, the engineer must now create a design. The engineer knows that the mean temperatures for June and July would be about 85°F and 90°F, respectively. Using these temperatures as input to the design model, evaporation estimates of 5.3 and 5.6 in., respectively, are obtained. Without further consideration, the engineer uses these estimates as the basis of the design. The high correlation of 0.98 provides the engineer with a false sense of comfort about the accuracy of estimates.

The problem with this description is that the engineer used the results of the analysis (the model) without understanding the analysis. If the hydrologist had known the engineer's objective, the hydrologist would have told the engineer that the model was based on data for the period from October to March, or if the engineer had inquired about the basis for the analysis, the engineer could have made an independent assessment of the results. For example, for the design location, the engineer could have compared the design evaporation estimates of 10.9 in. with the mean May–October lake evaporation of 50 in. reported in a climatic atlas. The estimate of 11 in. would then have been rejected as being unreasonably low.

Although this is a simplified example, it is not uncommon for a hydrologic model to be used without the user taking the time to determine the analysis that underlies the model. In cases where the user is fortunate enough to be applying the model within the proper bounds of the analysis, the accuracy of the design is probably within the limits established by the analysis; however, all too often inaccurate designs result because the assumptions used in the analysis are not valid for the particular design. The moral of the story is that those involved in the analysis phase should clearly define the limits of the model and those involved in synthesis or design should make sure that the design does not require use of the model outside the bounds established by the analysis.

1.4.2 A Conceptual Representation. Because of the importance of the concepts of analysis and synthesis, it is worthwhile to place the problem in a conceptual framework. We will conceptualize the hydrologic system to consist of three parts: the input, the output, and the transfer function. This conceptual framework is shown schematically in Figure 1.3. In the analysis phase, the input and output are known, and the analyst must find a rational model of the transfer function. When the analysis phase is completed, either the model of the transfer function or design tools developed from the model are ready to be used in the synthesis phase. In the synthesis or design phase, the design input and the model of the transfer function are known, and the predicted system output must be computed; the true system output is unknown.

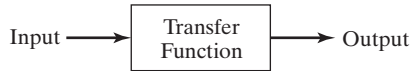


Figure 1.3 Conceptual representation of the system's process.

The designer predicts the response of the system using the model and bases the engineering-design solution of the predicted or synthesized response.

The reservoir-design problem can be used to illustrate this conceptual framework of hydrologic analysis and design. In this simplified problem, evaporation is needed for design; thus, evaporation serves as the output variable. Temperature serves as the input. In the analysis phase, the model for predicting evaporation is evaluated; therefore, the model represents all of the hydrologic processes involved in converting temperature (e.g., heat energy) into evaporation. Obviously, temperature is only a single measurement of the processes that effect evaporation, but it is a surrogate variable for these processes in the conceptual representation of the system. In the design phase, the designer uses other temperature conditions as input to the model to predict the response (e.g., the output) of the system. It is this synthesis of the system output that is used for the design. In summary, in the analysis phase, the temperature (input) and evaporation (output) were known, whereas in the synthesis phase, the temperature (input) and the model (transfer function) were known. Examples of the analysis/synthesis representation that is discussed in other chapters are given in Table 1.1.

EXERCISES

- 1.4.1** How do the words *analysis* and *synthesis* apply to a pipe system used to drain flood runoff from a housing development?
- 1.4.2** Outline a water budget for a small pond used by a farmer to irrigate crops and water livestock.

1.5 THE HYDROLOGIC BUDGET

The systems concept of Figure 1.3 can be applied to the elements of the hydrologic cycle, which can be viewed as inputs, outputs, and storages. Rainfall can be viewed as an input to the surface of Earth. The surface can be viewed as a series of storage elements, such as storage on the surface of vegetation and depression storage. Runoff from the surface can be viewed as an output from surface-storage elements, which would be a systems representation of the physical processes that control surface runoff.

If river channel processes are the important elements of hydrologic design, then surface runoff can be viewed as the input, the channel itself can be viewed as the storage element, and the runoff out of the channel (into another channel, a lake, or an ocean) can be viewed as the output from the system. Hydrologic analysts and designers make use of this systems representation of the elements of the hydrologic cycle to develop design methods to solve engineering problems.

Table 1.1 Examples of hydrologic analysis and synthesis

Input	Model or Transfer uncton	Output	Analysis		Synthesis	
			Known	Unknown	Known	Unknown
1. Runoff coefficient (C) Rainfall intensity (i) Drainage area (A)	Rational formula $Q_p = CiA$ (Equation 5.13) Figure 6.20	Peak discharge (Q_p)	i, A, Q_p	C	C, i, A	Q_p
2. Peak inflow rate (Q_i) Allowable outflow (Q_o) Runoff volume (V_r)		Volume of storage (V_s)	Q_i, Q_o, V_r, V_s	Figure 6.20	Q_i, Q_o, V_r	V_s
3. Upstream hydrograph (I)	Muskingum equation $O_2 = C_0I_2 + C_1I_1 + C_2O_1$ (Equation 8.26)	Downstream hydrograph (O_2)	I, O_2	C_0, C_1, C_2	I, C_0, C_1, C_2	O_2
4. Erosivity index (R) Soil-erodibility factor (K) Topographic factor (T) Cover factor (C) Support factor (P)	Universal soil-loss equation $E = RKTCP$ (Equation 12.14)	Soil loss (E)	E, R, K, T	C, P	R, K, T, C, P	E
5. Length (HL) Slope (Y) Storage (S)	NRCS lag equation $L = \frac{HL^{0.8}(S + 1)^{0.7}}{1900Y^{0.5}}$ (Equation 3.51)	Watershed time lag (L)	L, HL, Y, S	Coefficients of Equation 3.51	HL, Y, S	L
6. Storage (S) Rainfall (P)	$Q = \frac{(P - 0.2S)^2}{P + 0.8S}$ (Equation 5.38)	Runoff volume (Q)	Q, P	S	P, S	Q

The conceptual representation of hydrologic systems can be stated in mathematical terms. Letting I , O , S , and t denote the input, output, storage, and time, respectively, the following equation is known as the linear-storage equation:

$$I - O = \frac{ds}{dt} \quad (1.1)$$

The derivative on the right-hand side of Equation 1.1 can be approximated by the numerical equivalent $\Delta S/\Delta t$ when one wishes to examine the change in storage between two times, say t_2 and t_1 . In this case, Equation 1.1 becomes

$$I - O = \frac{\Delta S}{\Delta t} = \frac{S_2 - S_1}{t_2 - t_1} \quad (1.2)$$

in which S_2 and S_1 are the storages at times t_2 and t_1 , respectively.

Equation 1.1 can be illustrated using the flow in a river as the input and output. Assume that the inflow and outflow rates to a river reach of 1,500 ft in length are 40 ft³/s and 30 ft³/s, respectively. Therefore, the rate of change in storage during a 10-min period equals 10 ft³/s or 6,000 ft³. Since the inflow is greater than the outflow, the water-surface elevation will rise. Assuming an average width of 16 ft, the increased storage would cause the water-surface elevation to rise by 6,000 ft³/[16 ft(1,500 ft)] = 0.25 ft over the 10-min period.

The storage equation of Equation 1.1 can also be used for other types of hydrologic problems. Estimates of evaporation losses from a lake could be made by measuring all inputs, such as rainfall (I_1), inflow from streams (I_2), and groundwater inflow (I_3); all outputs, such as streamflow out of the lake (O_1), groundwater flow out of the lake (O_2), and evaporation from the lake (O_3); and the change in storage between two time periods, which could be evaluated using the lake levels measured at the beginning and end of the time period. In this case, both the input and the output involve multiple measurements. In other words, they are vector quantities with the elements of each vector I_i and O_j representing the different components of the inflow and outflow from the system. The unknown is one of the elements of the outflow vector. Mathematically, the water balance is

$$(I_1 + I_2 + I_3) - (O_1 + O_2 + O_3) = \frac{ds}{dt} \quad (1.3)$$

Estimating evaporation losses (O_3) would require measurements of each of the inflows, the other outflows, and lake levels; obviously, any errors in the measurements of the elements of Equation 1.3 would affect the accuracy of the estimated evaporation. Also, if there are any inflows other than I_1 , I_2 , and I_3 or if there are any outflows other than O_1 and O_2 , the estimated evaporation will be in error by such amounts.

In summary, the hydrologic budget is a convenient way of modeling the elements of the hydrologic cycle. It will be used frequently in describing the problems of analysis and design.

EXERCISES

- 1.5.1** If the inflow to a pond from a 52-acre watershed is 0.8 area-inches during a 12-hour storm and the mean outflow rate is $1.1 \text{ ft}^3/\text{s}$, what is the change (increase or decrease) in storage (acre-ft) over the duration of the storm?
- 1.5.2** During a 1.5-hour storm, 1.3 in. of rain fell during the first hour, and 0.4 in. of rain fell during the last half hour. The average flow in the downspouts was 1.7, 3.6, and $3.4 \text{ ft}^3/\text{s}$ for each of the half-hour periods. If the rooftop was 3 acres, how deep was the water on the roof at the end of each half-hour period?

1.6 INTRODUCTION TO PROFESSIONAL ETHICS

The chapters of this book detail various methods of hydrologic analysis and design, which may imply that the heart of hydrologic work is number crunching. While hydrologists must certainly understand physical processes and know design procedures, they must also recognize that designs are necessary because society has needs and problems. Very often, these needs and problems arise because of value issues. An adequate water supply is important for both public health and the happiness of citizens. But, a dam and reservoir may represent a hazard to public safety. The hydrologist will be involved in the technical aspects of solving the problem, but the hydrologist also has social responsibilities related to value issues. These value issues should be just as important to the hydrologist as the technical details. Sound hydrologic practice is not limited to providing correct evaluations of design elements. Hydrologists must be able to recognize value issues, consider the value issues along with the technical issues, and know how to incorporate these value issues into decision making. These value-based responsibilities are just as important as design responsibilities.

1.6.1 Case Studies: Ethical Dilemmas in a Professional Setting. To create a foundation for a discussion of the social responsibilities of professionals who are involved in hydrologic analysis and design, it is worthwhile to sketch four hypothetical situations that involve value issues in a professional setting. The initial emphasis should be on identifying the value issues. The task of solving value dilemmas that arise in professional practice is discussed later.

Case Study 1. The firm for which you work obtains a contract for designing a pipe to carry a liquid hazardous waste between the plant where it is produced as a by-product and a storage facility where it is held until it can be transported to a disposal site. The two sites are 1.2 miles apart. During the site investigation, you find that the route proposed by the client passes through a very permeable geologic formation (see Chapter 6). You recognize that a leak or rupture of the pipe in the area of the permeable formation could cause significant problems with the quality of the water withdrawn from the aquifer.

You bring the problem to the attention of your supervisor and propose an alternative route for the pipe. Your proposal would increase both the length of the

pipe and the cost of the project, as well as delay the completion date of the project. Your supervisor tells you that the project is to be completed as planned and that the potential hazard is not your concern. What values are in conflict? What action, if any, should you take?

Case Study 2. The firm that you work for obtains a contract for constructing a water-supply reservoir. You find out that the proposed site is above an area that is zoned for commercial and high-density residential land uses, although the areas are currently undeveloped. Based on this information, you recognize that state policy would require the hydrologic design to be based on a probable maximum flood (PMF) analysis (see Chapter 4). You indicate to your project manager that you have no experience with PMF analyses and that no one else in the firm has the appropriate level of experience. After you recommend that the firm hire an outside consultant who has the necessary experience, the project manager indicates that your proposal is not economically feasible and that you will be responsible for the design. What are the value issues involved in this case? Recognizing that the effort is outside your area of professional competency, what action, if any, should you take?

Case Study 3. A public agency issues a request for proposals on a design project that is in your area of professional expertise. Your supervisor indicates that you will be in charge of writing the proposal, which you believe will be a 3- to 4-week effort. Your supervisor tells you to charge the time to a project that is currently in progress, but to which you have no involvement, and that is totally unrelated to the project for which you will be writing the proposal. You believe that your time should be reimbursed using company-overhead funds rather than funds from another client. Your supervisor rejects your concern about charging time to a project on which you are not involved and states that if you want to get paid, you must charge your time as instructed. What value issues are in conflict? What action, if any, should you take?

Case Study 4. The firm that you work for discharges wastes into a local stream. It is your responsibility to take samples for specific water-quality indicators, perform the necessary laboratory tests, and report the results to the local public-health agency. On a regular basis, you inform your supervisor of the test results prior to providing a summary of the results to the local public-health agency. At one reporting date, you find that the effluent from your firm's plant does not meet state water-quality standards. When you present the results to your supervisor, she indicates that the test results are probably wrong and that you should send in a report to the public-health agency stating that the effluent meets the state water-quality standards for the reporting period. Failure to meet the standards can result in a fine or a plant process shutdown. What action should you take? Should you report the true test results to the public-health agency or falsify the report as ordered by your supervisor?

1.6.2 Case Studies: A Question of Values. The four case studies presented in Section 1.6.1 should not necessarily be viewed as a problem of employee versus employer. The heart of each of the dilemmas is the employee's struggle with his or her own value system. Before the employee further confronts the supervisor or project manager, the employee must evaluate the value conflicts inherent in

each situation. Such evaluation is a prerequisite to finding an optimal solution. The employee also must know what the profession and society expect of him or her.

To solve each of these dilemmas, the employee must address a number of concerns. These concerns can be summarized by the following four questions:

1. What are the value issues involved?
2. How important are the consequences associated with the issues?
3. How can the issues and consequences be weighted so that a decision can be reached?
4. What procedure should be followed to arrive at a solution?

These four questions reflect the theme of the following sections of this chapter.

1.7 THE DIMENSIONS OF PROFESSIONALISM

When we hear the word *professional*, we may think of a professional athlete or maybe even a hired killer. After all, the most general definition of the word *professional* provided by a dictionary would be something like the following: one who is engaged in a specific activity as a source of livelihood. Using this definition, almost all people who work could be classed as professionals. However, some professions, such as doctors, members of the clergy, and teachers, usually are thought to require a higher level of knowledge than others that just satisfy the dictionary definition of a professional because they serve society in special ways. It is in this sense that we want to classify hydrologists as professionals; while hydrologists engage in specific activities as a source of livelihood, their work carries special responsibilities to society that accord their work the status of a profession.

In his discussion of professional development, Dougherty (1961) defines a professional as

one who uses specialized knowledge and skill in the solution of problems that cannot be standardized. He [or she] is actuated by a service motive; he [or she] works in a relation of confidence, and observes an acceptable code of ethical conduct.

But, it must be emphasized that the term *professional* does not apply just to someone who has specialized knowledge and skill. Other dimensions of professionalism also are important. In the definition above, Dougherty recognized the multifaceted nature of professionalism, and from this he makes the following conclusion:

Professionalism is a way of thinking and living rather than an accumulation of knowledge and power. Knowledge and power are essential and when actuated by the professional spirit they produce the leaders and torch-bearers.

What are the fundamental dimensions of professionalism, all of which are necessary for an individual to be considered a professional? NSPE (1976) identifies five characteristics of a professional engineer that also apply to those who practice hydrologic design work:

1. One who possesses a body of distinctive technical knowledge and art gained by education, research, and experience.
2. One who recognizes a service motive to society in vital and honorable activities.
3. One who believes in standards of conduct, as represented by ethical rules.
4. One who supports the legal status of admission to the profession (the most common forms are registration to practice engineering and graduation from an accredited engineering curriculum).
5. One who takes pride in the profession and has a desire to promote technical knowledge and professional ideals.

To these five characteristics, Dougherty's definition identifies a sixth factor that should be emphasized as a fundamental characteristic, even though it may be inherent in the five characteristics listed.

6. One who works in a relation of confidence to his or her employer, the public, and all who use his or her works.

Considerable time, education, and professional practice are required to obtain the specialized knowledge and skill that characterize a professional. The professional, in contrast to the artisan or technician, is capable of synthesizing experience and fundamental principles in solving new problems. Such problem solving requires independent thought and motivation that is not self-serving. The motivation must stem from a dedication to service of both society and the profession; otherwise, the knowledge and skill do not maximize human welfare. Because of the interaction of a professional with society and the advantage that the professional has through his or her specialized knowledge and skill, it is important for a professional to have sound moral judgment and be able and willing to translate this moral judgment into principled professional conduct. While the six dimensions of professionalism are independent in that any person may possess one or a few of these characteristics, a true professional must possess all six of the fundamental dimensions.

EXERCISES

- 1.7.1 Six characteristics of professionalism are identified. Indicate how each one applies to a medical doctor.
- 1.7.2 How could an engineering student fulfill each of the six characteristics of professionalism?
- 1.7.3 How does copying another person's homework violate the concept of professionalism?

1.8 HUMAN VALUES: A DEFINITION

A *value* can be defined as a principle or quality considered worthwhile or desirable. As used here, a *human value* is a quality that relates directly to the potential for improving the conditions of humankind. The ability to make a value

decision is one of the characteristics that distinguish human beings from other living things. Along with this ability goes the responsibility to make value decisions in an unbiased and precise way. To make accurate value decisions, it is necessary for the decision maker to understand his or her value system and establish value priorities. He or she must also know and understand the values held important by the profession, which requires a certain commitment to examine critically one's own value priorities. Unfortunately, too few people take the time to make an assessment of their value system. Individuals often are guided by standards established by their family and friends; peer pressure is all too often used as an excuse for improper conduct.

While “life, liberty, and the pursuit of happiness” is probably the most famous value statement, and “I love you” is probably the most frequently used value statement, there are numerous other values. The following is a partial list of human values:

Beauty/aesthetics	Honesty	Mercy
Cleanliness	Honor	Pleasure/happiness
Courage	Kindness	Public health/safety
Diligence	Knowledge	Respect
Efficiency	Life	Security
Equality	Love	Truth
Freedom/liberty	Loyalty	Wisdom

These values represent feelings or attitudes rather than physical entities. Nonetheless, a scale could be developed in an attempt to reflect variations in a feeling or attitude. Industrial engineers are constantly trying to find rules and yardsticks to measure efficiency. Environmental engineers develop standards that define when minimum levels of public health are not being met. Engineering planners attempt to define indices that measure the beauty of an engineering project, and wildlife biologists develop indices to measure changes in the quality of life for animals whose habitat is affected by engineering projects. While such attempts make it possible to incorporate value issues into the quantitative decision-making framework, all too often hydrologists, as well as those in other professions, fail to recognize that everyone affected by a project may not agree with the scale used to quantify a human value. Such disagreement is the basis for value conflicts in a professional setting. However, at least the hydrologist can attempt to incorporate value issues into the decision-making process. The relative effort that a hydrologist expends in developing a scale to measure the various states of a value issue may be an indication of the sensitivity of the hydrologist to human values.

EXERCISES

- 1.8.1 Provide dictionary definitions of the following values: loyalty, respect, and honesty.
- 1.8.2 Distinguish between a human value and a professional value.

1.9 VALUES IN HYDROLOGY

While all hydrologists recognize the importance of technical issues in the workplace, they should also recognize that hydrologists do not practice in a value vacuum. Decisions involving human values are important to the individual, the firm, the profession, and society. In making value-based decisions, the individual usually places far greater weight on the consequences to himself or herself, such as the potential loss of employment or stature within the professional community, than on the consequences to the firm, the profession, or society. Also, the individual quickly becomes aware of the personal stress that accompanies value conflicts. The individual has moral obligations to his or her family, and these personal responsibilities cannot be ignored when solving ethical dilemmas in the workplace. Although professional codes of ethics do not require the hydrologist to be self-sacrificing, the individual must be willing to consider values important to the firm, the profession, and society in making value decisions, not just the values that are important to him or herself. Proper value decision making requires the individual to give weight to and properly balance the value responsibilities of each of the parties.

To make value decisions that best meet the ethical rights of all parties involved, the values that are important to all parties need to be identified and respected. In the case of the hydrologist who was told to file false time reports, he or she placed emphasis on values such as honesty and security. He or she may not have recognized the values that were important to his or her supervisor. The supervisor, in turn, placed emphasis on values such as loyalty, performance, and respect but showed little regard to the values that were important to either the hydrologist or society. Neither the hydrologist nor the supervisor recognized that their actions had an impact on values that are important to the profession, such as honor, and to society, such as truth in matters that affect society. Such failure to recognize values that are important to all parties involved may lead to nonoptimal decisions. In such cases, incorrect value decisions can lead to expulsion from the profession.

Values that should be important to the firm include honesty, responsibility, loyalty, efficiency, and harmony. Failure of an individual hydrologist to consider the value rights of the firm in solving ethical conflicts reduces the effectiveness of the firm in meeting its social responsibilities. A person's failure to recognize the values that are important to the firm is often viewed by the firm as a reflection of disloyalty.

While ethical problems such as those discussed in Section 1.6.1 may appear to have effects only on the person and the firm, they, in fact, can have an impact on the profession and society. Knowledge, honor, and integrity are just three human values that are important to the profession, and incorrect decisions regarding an ethical conflict can adversely affect these values. For example, the use of computer software for design by a hydrologist who is not familiar with the underlying theory might imply to society that knowledge, as well as the social value of public safety, is not important to the profession. In this case, the design hydrologist and the firm appear to be insensitive to values that are considered to be important to both the profession and society. Failed designs can lead to unnecessary regulation of the

profession by social institutions. Society places a high value on public safety and efficiency, with the value of efficiency being interpreted as human welfare and the optimum use of public resources. If a number of hydrologic designs were to fail because hydrologists were using software for designs outside their area of technical competency, society's image of both the firms and the profession would be tarnished.

The practice of hydrology involves more than just technical decision making. No quantifiable value goals are just as important as the benefit–cost ratios that reflect quantifiable monetary goals. Therefore, value decision making is important both within the firm and in the actions of hydrologists in their relations with the public.

From this discussion, it should be evident why value decision making in one's professional life is more complex than many of the value decisions made in one's personal life. When making a personal value decision, just one or two parties are usually involved (e.g., oneself and a friend). When making value decisions in professional life, numerous parties are involved. In addition to ethical responsibilities to oneself and family, the professional has legitimate and important ethical responsibilities to the employer, the client, the profession, and society. Failure to recognize and give proper consideration to each of these responsibilities can lead to nonoptimum decisions and even the appearance of professional misconduct.

1.10 SOLVING VALUE DILEMMAS

What should a young hydrologist do when instructed to falsify time reports or reports on water-quality test results? What should the hydrologist who is responsible for designing a pipe to carry a hazardous waste do when told not to worry about failure of the pipe? It should be evident that the mishandling of an ethical dilemma can have adverse effects on the individual, both professionally and personally; furthermore, there may be negative effects on the firm, the profession, and society. On the other hand, following the proper procedure may actually result in many human value benefits.

The most important element in solving ethical conflicts within the workplace is to distinguish between internal and external appeal. *Internal appeal* refers to actions taken within the firm to solve an ethical dilemma. *External appeal* refers to actions with respect to parties outside the firm. Except under unusual circumstances, all internal options should be considered in attempting to resolve an ethical conflict before seeking an external solution. The internal options are separated into three levels (see Table 1.2): individual preparation, communication with the supervisor, and initiation of the formal internal-appeals process. It is important to follow the appeals process in the sequence shown. As a first step, it is to the advantage of the employee to develop accurate records of the facts and details of the conflict, familiarize himself or herself with the appeals process for resolving ethical conflicts within the company, and identify alternative courses of action. At this point, it is also extremely important for the individual to know exactly what

Table 1.2 Procedure for solving ethical conflicts

I. Internal appeal options
A. Begin individual preparation
1. Maintain a record of events and details
2. Examine the firm's internal appeals process
3. Know federal and state laws
4. Identify alternative courses of action
5. Specify the outcome that you expect the appeal to accomplish
B. Communicate with your immediate supervisor
1. Initiate an informal discussion
2. Make a formal appeal
3. Indicate that you intend to begin the firm's internal process of appeal
C. Initiate an appeal through the internal chain of command
1. Maintain a formal communication appeal
2. Formally inform the company that you intend to pursue an external solution
II. External appeal options
A. Undertake personal options
1. Engage legal counsel
2. Contact a professional society
B. Communicate with the client
C. Contact the media

he or she expects any appeal to accomplish. It is inadequate just to state that there appears to be a problem; ideally, the employee can propose a solution that can resolve the problem. As stated previously, the individual must recognize the moral rights of the firm, the profession, and society instead of just showing concern for the effects on himself or herself.

Once the individual has studied and documented the facts and formulated a plan for the appeal, the matter should be informally discussed with the immediate supervisor. In most cases, problems can be resolved at this level, which is to the benefit of all parties involved. In some cases, a more formal appeal to the immediate supervisor is necessary. This appeal is usually in the form of a memorandum that clearly states the facts, the individual's concerns, and the actions that would be necessary to resolve the concerns. It should be evident that good communication skills are important in resolving ethical conflicts because a failure to clearly state the concerns, whether orally or in writing, may hinder the resolution of the problem. Appealing the problem to higher levels before completing the appeal to the immediate supervisor is viewed negatively by all involved and decreases the likelihood of a favorable resolution. Finally, before continuing the appeal beyond the immediate supervisor, an individual should inform the supervisor in writing of the intention to continue the appeal to higher authorities.

The process of appealing an ethical conflict within the firm is usually quite similar to the interaction with the immediate supervisor. Formal steps should follow informal discussions, and steps within the process should not be bypassed. If the internal appeals process is completed without resolution of the conflict, the

individual should formally notify the company that he or she intends to continue with an external review of the problem.

Before expressing any concern to the client or making concerns public, a lawyer and/or professional society should be consulted. Lawyers can identify courses of action and pitfalls in external appeal. Whereas lawyers understand legal constraints and consequences, they often lack the technical expertise necessary to evaluate the technical adequacy of arguments. Professional societies have such technical expertise available and may be willing to informally discuss the case and make recommendations; however, they may not have a complete understanding of the legal implications.

After engaging legal counsel and/or obtaining advice from a professional society, the individual may elect to approach the client. In many cases, the client lacks the expertise to judge the accuracy of the claims and will rely on the advice of the firm in evaluating the concerns; however, if the issues involved in the conflict can be accurately communicated, the client may pressure the firm to resolve the issue internally. Alternatively, the client may expend the resources necessary to obtain an unbiased review of the matter. While it is extremely rare that it will be necessary to carry the appeals process to the client, the effects on one's professional career can be detrimental if the results fail to validate the arguments.

The last resort in resolving ethical conflicts is public disclosure. The benefits and costs in terms of both economic criteria and nonquantifiable value goals should be assessed rigorously before the media is contacted. It is not in the best interest of society to make public disclosure prior to seeking an internal resolution. The only case where public disclosure is acceptable prior to following the appeals process (Table 1.2) is when immediate danger to the public exists.

Four ethical dilemmas were introduced in Section 1.6.1. While the four situations were quite different in many respects, the process shown in Table 1.2 can be used in resolving each of the conflicts. The problem of falsifying time reports is best handled with the hydrologist informally discussing the situation with the supervisor. In the discussion, the hydrologist should clearly state the value conflict created by the supervisor's request to falsify the timesheets. He or she might try to persuade the supervisor to change the way in which project accounts are handled so that actions that violate an employee's value system are not necessary. Furthermore, the employee might point out that such a requirement reduces the employee's respect for the firm and may affect his or her sense of loyalty. If the hydrologist appeals to the supervisor in terms of the values that are of interest to the firm, the supervisor may be more responsive than if the hydrologist only showed concern about his or her own position.

EXERCISES

- 1.10.1** Explain why it is wrong to seek an external solution to an ethical problem prior to exhausting all efforts to achieve an internal solution? Are there any circumstances where an internal solution can be bypassed? Explain your response.
- 1.10.2** How might a company's code of conduct be useful in achieving an internal solution to an ethical problem within the company?

1.11 RATIONALIZATION

In Case Study 4 of Section 1.6.1, how does the supervisor justify asking the employee to falsify the water-quality report? In Case Study 2, how does the project manager justify asking someone to perform hydrologic-engineering work that is clearly outside the individual's competency? If instead we were to ask how a student justifies cheating on a test, the response would probably involve "rationalization." In many cases, rationalization is a primary factor in unethical professional conduct.

Rationalization is the mental process of completely falsifying the results of one's actions so that he or she is incapable of regretting the action (Ross, 1972). It is a form of self-deception for the purpose of justifying one's actions. Rationalization serves two purposes. First, it serves as an excuse to avoid guilt feelings and the scorn of others. Second, it provides self-justification of an action, thus excusing the individual from making a change in his or her moral decision process (Ross, 1972).

What possible rationalizations could professionals employ to provide self-justification for their actions? In Case Study 4, the supervisor justifies the action of requesting the engineer to file a false report by arguing that the test results are probably incorrect and that, if additional tests were made, they would prove that the standards are being met. The supervisor believes that his or her professional experience is more accurate than some laboratory tests.

Other common rationalizations to justify unethical conduct include "everybody is doing it" and "if I don't do it, someone will." If individuals can convince themselves that "everyone is doing it," then they feel less guilt because they believe that they are on the same moral level as others in comparable positions. Using such logic, it follows that it is not necessary to rise to a level of ethical conduct that is higher than that followed by one's peers. Rationalization makes it unnecessary to consider the value rights of others and enables the supervisor to repeat the conduct in the future.

What rationalization might the young engineer make in order to charge the time to the incorrect account? The following are a few responses:

1. It is standard business practice.
2. I am only doing what I am told to do.
3. It is only a small amount of money.
4. It is the manager's responsibility to decide money matters.
5. I want to be loyal to the company.

Any one of these responses might serve the immediate needs of the individual and the firm but do these responses meet the needs of the profession, society, or the client? Society values truth. The profession values the integrity of its members, as well as its reputation. The client values trust. These values are being given zero weight in the decision. Instead, the young engineer is placing value on personal needs, and the supervisor is valuing efficiency within the firm. Neither the engineer nor the supervisor is properly balancing value issues. Their rationalizations are serving as the means to place selfish ends before the legitimate expectations of the profession and society.

EXERCISES

- 1.11.1** Does rationalization represent a dishonest act? Explain your response.
- 1.11.2** If a person rationalizes an unethical action, would this rationalization influence the individual's response to similar ethical dilemmas? Explain your response.

1.12 CONCLUDING THOUGHTS

Hydrologists do not work in a value vacuum, and it is unreasonable to believe that you will never be confronted by a value conflict in the workplace. If appeals are handled properly, ethical conflicts will usually be resolved without incident. The resolution of more complex ethical conflicts requires a good understanding of the systematic process given in Table 1.2, which is designed to give the optimum weight to the moral rights of all the parties involved. Resolution of ethical conflicts should not be viewed solely from the viewpoint of the effects on the individual. Just as the firm, the profession, and society have moral responsibilities, they also have value rights that must be respected. Proper respect for these values is in the best interest of society as well as the individual, and the responsibilities are best met when a procedure such as that outlined in Table 1.2 is followed.

Solutions to ethical dilemmas require knowledge of three aspects of ethical decision making. First, the person involved must have an appreciation for human-value issues and must be able to identify the values that are in conflict. Second, the person must recognize that the effects of value conflicts extend beyond himself or herself. Value conflicts within the workplace have an impact on the firm, the profession, and society, not just on the individual. Further, the individual, the firm, the profession, and society have ethical responsibilities as well as ethical rights. Moral rights and responsibilities to each of these parties must be considered in finding the optimal solution to an ethical dilemma. Third, the individual must know the steps to follow that will provide the optimum solution to an ethical dilemma. These steps are summarized in Table 1.2.

The discussion in this chapter has tried to show that the hydrologist has both technical and value professional responsibilities, both of which require the hydrologist to show concern for the public. While I may have ignored the employer, the hydrologist also has a direct responsibility to the employer, and this responsibility goes beyond just fulfilling the technical requirements of the job. Important employer-related responsibilities must be considered in decision making. Before going beyond the firm to protest corporate actions, the hydrologist should make every attempt to view the problem and the value issues from the perspective of the employer. The professional reputation of both the firm and its other employees should be important to the hydrologist even when he or she disagrees with the employer's decision. The hydrologist should also respect the collective knowledge and experience of the managers of the firm. The hydrologist should recognize that the employer has a responsibility to provide long-term employment for the

employees of the firm. I personally know of many employers and managers in engineering firms who place high priority on the welfare of the employees and personally agonize over decisions to discharge employees because the funding is no longer there to support them. These are important value issues that increase the complexity of a decision to protest the action of an employer.

The main point of this discussion is that hydrologic design requires more than just technical knowledge of design methods. Human and social values must be considered in the practice of hydrology, which should be quite evident from the four case studies presented in Section 1.6.1. While technical details are often standardized, it is rarely possible to standardize value issues. Value issues cannot be easily quantified and numerically combined with quantifiable goals such as maximizing benefit–cost ratios. Our inability to quantify these value issues does not mean that we should not incorporate them into our decisions. They must be considered—otherwise, the hydrologist will be viewed as lacking a social conscience. In the first case study of Section 1.6.1, how can we quantify the increased risk to the health of the local citizens? In deciding on a course of action in the fourth case study, can the damage to the environment and the potential reduction in the aesthetic qualities of the riverine environment be quantified with the cost of cleanup?

I have been asked on numerous occasions for advice on how to handle ethical dilemmas. In the initial stage of one such problem, I discussed the issue with both parties and thought that I could resolve the issue. Therefore, I ignored Step IA1 of Table 1.2. Instead of making notes after each of the discussions, I mistakenly assumed that the situation was being resolved. I made a written statement only after I was so instructed to do so by a lawyer who was approached by one of the parties. At that point, I could not remember all of the details and left out what turned out to be some very important events. In retrospect, it may have hindered my ability to help the individual and resolve the issue. Learn from my experience. When a problem arises, no matter how minor it seems at the time, maintain a record of it. The details could prove to be very valuable at a later time. A tape recorder is a very efficient means of recording the details of events. However, whatever medium is used to record the information, make sure that it is stored in a secure environment.

In summary, every student of hydrology must recognize that the study of hydrology involves more than just developing skills related to the technical aspects of hydrologic design. The hydrologist must also consider the social impact of a design.

PROBLEMS

Problem 1.1

Identify the processes of the hydrologic cycle that affect flood runoff from a 3-acre forested lot. Discuss the relative importance of each of the processes. If the lot is cleared, what changes in the processes and their importance will occur? If a single-family residence is constructed on the cleared lot, what processes will control flood runoff from the lot?

Problem 1.2

If the drainage area of the site of Problem 1.1 is 50 acres and 150 residences are to be built, identify the processes of the hydrologic cycle that will affect flood runoff for the following three watershed states: forested, cleared, and developed. In each case, discuss the relative importance of the processes.

Problem 1.3

If a 25-acre commercial shopping center is to be constructed at one end of a 50-acre watershed, identify the processes of the hydrologic cycle that will affect flood runoff after development. Discuss the relative importance of the processes.

Problem 1.4

Would you expect a 25-acre commercial shopping center to affect the flood runoff from a 500-acre watershed, assuming that the remainder of the watershed is forested? Discuss your responses in terms of the hydrologic cycle.

Problem 1.5

Identify the hydrologic processes that affect flood runoff from a 3,000-ft section of interstate highway and the accompanying grass-covered right-of-way. Which processes are most important in determining the flood runoff from the site?

Problem 1.6

A wet basement (e.g., moisture on the floor and water seeping in through cracks in the wall and floor) is not an uncommon problem in single-family residential homes. From the standpoint of the hydrologic cycle, briefly discuss the problem and possible solutions.

Problem 1.7

A single-family residential development lies down slope from a small suburban shopping center that has no facilities for controlling flood runoff. Several of the homes experience erosion of the lawns during moderate and large storm events. From the standpoint of the hydrologic cycle, briefly discuss the problem and possible solutions.

Problem 1.8

For a circular pipe flowing full, the head loss (h_L) can be computed using the Darcy–Weisbach equation,

$$h_L = f \left(\frac{L}{D} \right) \frac{V^2}{2g}$$

in which f is the dimensionless friction factor that is a function of the Reynolds Number and pipe roughness, L and D are the length and diameter of the pipe, respectively, and V is the velocity of flow. A firm intends to develop a pipe made of a new ceramic material for which f values are not available. Discuss the task in terms of analysis and synthesis.

Problem 1.9

Design an experiment in which the efficiency of a pump at some specified Q can be determined. Discuss the problem in terms of the concepts of analysis and synthesis.

Problem 1.10

A storm with a uniform depth of 1 in. falls on a 25-acre watershed. Determine the total volume of rainfall in acre-feet and cubic feet. If all of the water were collected in a storage basin with vertical walls and a floor area of 25,000 ft², how deep of a basin would be needed?

Problem 1.11

A hurricane has an average rainfall depth of 9 in. over a 200-mi² watershed. What size reservoir in acre-ft would be required to contain 20% of the rain completely?

Problem 1.12

The following table gives the measured rainfall and surface runoff from a 65-acre watershed. Values given are averages over the time interval. Compute the accumulated storage (ft³) of water within the watershed. Plot on the same graph the inflow, outflow, and cumulative storage as a function of time.

Time (min)	Rainfall (in.)	Runoff (ft ³ /s)
0	0	0
10	0.07	2.1
20	0.15	9.5
30	0.21	18.9
40	0.16	37.3
50	0.09	48.8
60	0.05	52.6

Problem 1.13

For the six dimensions of professionalism given in Section 1.7, discuss how each applies to the practicing hydrologist. Provide examples to illustrate each point.

Problem 1.14

Discuss the importance to the professional hydrologist of maintaining professional competence, including the importance to the individual, the firm, the profession, the client, and society. Also, discuss ways that the hydrologist can maintain professional competence.

Problem 1.15

Select five values from the list provided in Section 1.8 and discuss how each can be important to society. Provide examples from hydrologic practice to illustrate each point.

Problem 1.16

Obtain a copy of the code of ethics for a professional society to which hydrologists might belong. Evaluate the human-values basis that underlies each section of the code.

For each of the examples given in Problems 1.17 to 1.23, provide an evaluation based on the four questions given at the end of Section 1.6.2. Use Table 1.2 to arrive at a recommended course of action.

Problem 1.17

The manager of a firm that performs hydrologic design work underbids on a contract in order to improve chances of being awarded the contract, knowing that he will later submit requests for contract “add-ons” with higher-than-reasonable budget amendments.

Problem 1.18

A hydrologist makes extensive personal use of the company’s computer system without approval or any expectation of reimbursing the firm.

Problem 1.19

A hydrologist, who is also a registered professional engineer, signs design work completed by unregistered subordinates without checking the work for accuracy.

Problem 1.20

A research hydrologist writes a paper that details personal research and submits the paper to two different professional journals without informing them of the dual submission. The paper is subsequently published in both journals.

Problem 1.21

A hydrologist, who serves as office manager for a large engineering firm, hires two new hydrologists, one man and one woman; both are recent college graduates with no experience. While both show equal ability and effort, the manager continually gives better assignments to the man and limits the professional opportunities of the woman.

Problem 1.22

A college professor, who has a full-time position teaching hydrology at a state university, spends almost all of her time at the firm where she serves as a consulting hydrologist. While she makes most of her classes, she greatly exceeds the university norm of “one day per week of outside consulting.”

Problem 1.23

A hydrologist whose only professional experience is in surface-water hydrology purchases computer software that can be used for solving problems in groundwater hydrology. The hydrologist then uses the software while selling his services as a consulting groundwater hydrologist.

Problem 1.24

For each of the cases in Problems 1.17 through 1.23, provide a rationalization that the individual might use to justify the improper conduct.

Problem 1.25

Give three (or more) rationalizations used by students who cheat on examinations. For each rationalization, show the irrationality of the reasoning.

Problem 1.26

Are students who cheat on tests and falsify lab results while in college more likely to participate in unprofessional conduct after they have graduated and when they are working in professional life? Support your response.

Problem 1.27

Critique the following statement of an engineering student: "I'm ethical, so I don't need to hear any lectures on professional conduct."

Problem 1.28

A woman actively participated in a research project while she was a student. Two years later, she realized that her advisor published the research results but did not include her as a co-author. What action, if any, should she take?

REVIEW QUESTIONS

- 1.1** Which one of the following tasks would not be an element of engineering hydrology? (a) Evaluate the effect of the size (capacity) of the outlet from a reservoir on recreation facilities at the reservoir site; (b) estimate the size of a pipe that would be required to carry floodwater under a highway; (c) evaluate the effect of water supply on the distribution and movement of people in the United States; (d) estimate the rates of storm runoff from rooftops.
- 1.2** Which one of the following is not considered part of hydrologic design? (a) The spatial variation of storm rainfall; (b) the economic value of the water; (c) the quality of the water supply; (d) the temporal variation of rainfall; (e) all of the above are considered.
- 1.3** Which one of the following is not a major factor that contributes to uncertainty in hydrologic designs? (a) The spatial and temporal variability of rainfall; (b) the effect of the soil and vegetation in converting rainfall to flood runoff; (c) the storage of flood runoff in river channels and floodplains; (d) our inability to accurately forecast future occurrences of rainfall; (e) all of the above are important factors.
- 1.4** Which one of the following elements of the hydrologic cycle is not affected by urbanization? (a) Rainfall; (b) depression storage; (c) interception storage; (d) channel storage; (e) all of the above are affected.
- 1.5** Consider a small forested area as a system. The rate of rainfall is measured with a raingauge in an open area about 100 ft from the forest. A second gauge that measures

the depth of rain that passes through the tree canopy is located under one of the trees near the center of the forest. In a problem of analysis, the objective would be to estimate (a) the rate of evaporation from the leaves of the tree; (b) the portion of the rain that is available for infiltration; (c) the water available for runoff; (d) the volume of interception storage.

- 1.6** If water travels at 6 ft/s through a 24-in. diameter culvert that passes under a 30-ft roadway, the travel time in minutes of a particle of water as it passes through the culvert is (a) 0.1; (b) 1; (c) 6; (d) 120.
- 1.7** For the data shown in Review Question 1.6, what is the volume rate of flow (ft^3/s) that passes through the culvert when it flows full? (a) Unable to be computed given the information provided; (b) 18.3; (c) 75.4; (d) 274.
- 1.8** Floodwater with an average depth of 1 in. over a 96-acre parking lot drains into a small river during a storm of 10 min. The average rate of water discharging from the river during the storm is $360 \text{ ft}^3/\text{s}$. The volume (in acre-ft) of water in the river will increase (+) or decrease (−) by (a) −7.5; (b) 3; (c) 7.5; (d) 96.



Note the edge of the storm cell.

2 Precipitation

CHAPTER OBJECTIVES

- Introduce the important characteristics of rainfall.
- Provide computational methods for estimating both average areal rainfall amounts and missing rainfall depths.
- Provide a method for evaluating the consistency of a gauged record of rainfall amounts.
- Introduce the concept of a temporally distributed design storm, which is an input to many hydrologic design problems.

2.0 NOTATION

A_s = storm area size	$\bar{P}$ = weighted average rainfall depth
A_x = average annual rainfall at gauge x	S_i = slope of part i of double-mass curve
d_i = distance between gauge i and regional center point	T = exceedance frequency (return period; years)
D = storm duration	w_i = weight applied to rainfall depth at gauge i
i = rainfall intensity (in./hr)	y_i = adjusted rainfall for section i of double-mass curve
n = number of raingauge stations (sample size)	ΔX_i = difference in regional catch for section i of double-mass curve
p = exceedance probability	ΔY_i = difference in catch at base station for section i of double-mass curve
$\bar{p}$ = average proportion	
P = estimated mean areal rainfall	
P_i = rainfall depth (in.) at gauge i	
P_x = missing rainfall not measured at gauge x	

2.1 INTRODUCTION

Precipitation can take many forms, including rain, snow, sleet, hail, and mist. With respect to hydrologic design, only rain and snow are important. This chapter is concerned with rainfall. The problem of snowmelt runoff is addressed in another chapter.

Precipitation is the water that falls either as a solid or as a liquid. As an air mass cools, moisture in the atmosphere condenses and falls. Rain, the liquid form of precipitation, is the water source for flooding. Snow, the solid form of precipitation, is also responsible for flooding when the snow melts. In some portions of the world, snowmelt runoff is the determinant of the annual maximum flood.

Three major types of storms are the cyclonic, convective, and orographic. The hurricane and the tropical cyclone are special cases of the cyclonic storm type. Heat given off by the Earth causes air masses to rise. These masses can contain large volumes of water and as the mass cools, the water condenses and falls. Rapid condensation of large masses causes large volumes of rain, which produces the major thunderstorms that contribute to large runoff events.

Hurricanes and cyclones generally develop over the ocean where water surface temperatures can exceed 85°F. The cyclonic storm tends to be the most severe when the air masses have the greatest temperature differential. They often produce rainfalls from 8 to 20 in. over periods of 24 hours, with winds reaching 75 mi/hr and much more. These extreme events are responsible for considerable volumes of rainfall and can greatly influence the shape and magnitudes of intensity-duration-frequency curves, which are a central element in hydrologic design.

Precipitation that occurs when air is forced over a mountain is referred to as orographic precipitation. Mountain slopes that are subject to large masses of moist air experience considerably more precipitation than the leeward sides of the mountains because the air is too heavy to flow over the mountain. Instead, the water drops out on the front side, which leaves the leeward side with considerably less precipitation. This is evident in the type and amount of vegetation.